

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Either Gas or radiation force/pressure and gravitational force / gravity (1) are balanced / equal and opposite / are in equilibrium (1) Or Outward force and inward force (1) are balanced / equal (1)	2			2		
		(ii)		Light from Spica will take 261 years to reach Earth (or v.v.) / this is the distance light travels in 261 years	1			1		
	(b)			Indicative content: Properties 1. Remainder of cycle: red giant → white dwarf... 2. Becomes larger, lower temperature / redder and brighter → smaller, higher temperature / whiter and dimmer. Reactions 3. Hydrogen fusion stops in centre of Sun and helium fusion starts – linked to red giant. 4. Helium fusion stops – no more fusion – linked to white dwarf. Forces 5. Red giant: Gas and radiation pressure in the core increases – linked to helium fusion – giving unbalanced forces and star expands 6. White dwarf: Gas and radiation pressure decreases when fusion stops causing star to shrink 7. End of red giant phase – outward forces cause outer layers to be blown away, contributing to a <i>planetary nebula</i> , leaving hotter core Other relevant points could be made – contributing to reactions or forces, e.g. When H-fusion stops the core shrinks and heats up until a new balance of forces is established	3	3		6		